

A Regime-Aware Model for Loan-Level Probability of Default

Vincent Fiifi Obbeng

Department of Mathematics, Kwame Nkrumah University of Science and Technology, Ghana

Supervisors: Dr. Maxwell Akwasi Boateng and Dr. Yao Elikem Ayekple

Introduction

1.1 Background and Problem

Loan-level probability of default (PD) is the conditional probability that a particular loan enters a defined default state within a specified horizon, given the information available at the prediction date. This study uses a loan-month prediction unit and a twelve-month serious-delinquency outcome. For a loan i observed in month t , the outcome is

$$Y_{i,t}^{(12)} = \mathbb{I}\left\{\max_{h=1,\dots,12} D_{i,t+h} \geq 90\right\}, \quad (1.1)$$

where $D_{i,t+h}$ is days past due and the loan has not already reached 90 days past due at month t . The corresponding PD is

$$\pi_{i,t}^{(12)} = \Pr\left(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t\right), \quad (1.2)$$

where $\mathcal{F}_t$ contains information available by month t . The event is therefore the first transition to 90 or more days past due during months $t + 1$ to $t + 12$. This threshold is an operational measure of serious mortgage delinquency and is consistent with regulatory treatment of material arrears as evidence of default (Basel Committee on Banking Supervision, 2004; Board of Governors of the Federal Reserve System, 2006).

Loan-level PD models must estimate probability levels as well as rank loans by risk. Discrimination measures whether higher-risk loans tend to receive higher scores. Calibration

measures agreement between predicted probabilities and observed event frequencies. Both properties matter because a well-ranked score can still overstate or understate the level of risk (Brier, 1950; Gneiting and Raftery, 2007; Van Calster et al., 2019).

Logistic regression is a standard loan-level credit-scoring method. It provides an explicit probability function and coefficients with a direct log-odds interpretation (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). A borrower-only specification, however, assumes that the relationship between borrower characteristics and default remains stable through calendar time. This restriction can be material when common economic conditions change the consequences of leverage, payment burden, interest rates, or recent delinquency.

Observed macroeconomic variables can be added to the logistic equation. Such models connect loan-level risk to the credit cycle and are useful in stress analysis (Bellotti and Crook, 2009, 2013). A purely additive specification still assumes that macroeconomic variables shift default risk without changing borrower sensitivities. It also represents a persistent and multidimensional economic state as a linear combination of contemporaneous indicators. These restrictions may be inadequate when the same borrower characteristic carries different risk under benign and stressed conditions.

1.2 Research Gap and Aim

The literature contains established loan-level scorecards, macro-augmented credit models, and latent-state credit-risk models. Their intersection is narrower. Borrower-only models often impose constant coefficients. Additive macroeconomic models do not allow the effect of borrower information to vary with the economic state. Regime-switching applications frequently address rating migrations, default intensities, or portfolio-level quantities rather than an interpretable loan-month PD (Bangia et al., 2002; Yu et al., 2018). This leaves an empirical question about whether a filtered macroeconomic regime probability can improve a transparent loan-level model through selected borrower–regime interactions.

The study aimed to develop and evaluate a regime-aware model for twelve-month mortgage PD. A two-state Gaussian hidden Markov model (HMM) reduced a set of public macroeconomic indicators to a filtered probability of stress. That probability entered a regularised logistic model directly and through selected interactions with borrower, loan, and payment-history information.

1.3 Objectives and Research Questions

The completed study had three objectives:

1. It estimated macroeconomic regime probabilities using a two-state HMM.
2. It developed a regime-aware loan-level PD model that combined borrower information with the estimated regime probabilities.
3. It evaluated the regime-aware model against borrower-only and macro-augmented logistic baselines.

The corresponding research questions are:

1. What macroeconomic stress probabilities did the two-state HMM assign over the study period?
2. How did the inclusion of the stress probability and borrower–regime interactions alter the loan-level PD specification?
3. How did the regime-aware model compare with borrower-only and macro-augmented logistic baselines on calibration and discrimination?

1.4 Contribution and Scope

The methodological contribution is an interpretable link between a two-state macroeconomic HMM and a regularised loan-level logistic model. The regime probability is a continuous in-

put. Selected interactions allow specified borrower effects to vary with that probability. The empirical contribution is an application to Freddie Mac mortgage origination and monthly performance data matched to public macroeconomic series. The evaluative contribution is a chronological comparison based on the Brier score, log loss, calibration diagnostics, ROC-AUC, PR-AUC, and monthly default-count error.

These contributions are limited in scope. The study does not claim that regime switching is new in credit risk. It tests a specific interpretable specification on one mortgage dataset and one terminal test period. The HMM parameters and factor transformations were estimated over the full study window, although the reported state probabilities were generated by forward filtering. The test-period stress probability was also nearly saturated. These features restrict real-time and cross-regime interpretation.

Chapter 2 reviews the evidence directly relevant to the model. Chapter 3 defines the data, HMM, logistic specifications, validation design, and evaluation measures. Chapter 4 reports the retained empirical results, answers the research questions, and states the main limitations.

Literature Review

Probability of default is conditional on an event definition, a horizon, a risk set, and an information set. These elements determine the estimand. In a loan-month model, the relevant quantity is $\pi_{i,t}^{(12)} = \Pr(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t)$, where the outcome records whether loan i first reaches serious delinquency during the following twelve months. A change in the delinquency threshold, prediction horizon, treatment of terminated loans, or information available at month t changes the probability being estimated. PD should therefore be interpreted as a forecast for an eligible loan at a defined date rather than as an unconditional borrower attribute.

This conditional interpretation separates PD estimation from the retrospective description of defaulted loans. A retrospective analysis asks which characteristics are common among realised defaults. A PD model asks how those characteristics alter the probability of a future event among loans that remain eligible to default. The risk-set restriction is part of that question. Including observations that are already seriously delinquent changes the task from transition prediction to state classification. Treating loans with incomplete follow-up as non-events changes the observed event frequency and can bias calibration. Loan-level PD estimation therefore requires the event construction and the eligible population to be stated together.

The horizon also affects the meaning of the probability. A one-month delinquency probability and a twelve-month serious-delinquency probability are not interchangeable. The longer horizon aggregates the chance of an event over several future months and generally

produces a higher cumulative event rate. It also increases the importance of adequate follow-up. The twelve-month horizon used in this study is fixed for every loan-month, which permits a direct comparison between predicted values and realised frequencies. It does not estimate a lifetime default probability or a time-to-event hazard.

Logistic regression remains a central method for loan-level credit scoring. It maps a linear predictor to the unit interval and provides coefficients that describe changes in log-odds. Credit-scoring research has used logistic models for both origination characteristics and behavioural information observed after account opening (Hand and Henley, 1997; Thomas, 2000; Costa e Silva et al., 2020). Origination variables describe initial credit quality, leverage, balance, interest rate, term, and borrower structure. Monthly performance variables describe the subsequent payment path. Recent delinquency, its duration, and evidence of cure can contain information that is not available in the origination record.

The logistic link provides a coherent probability only when the fitted linear predictor is combined with an appropriate estimation and calibration procedure. The coefficient on a continuous variable describes a change in log-odds holding the other included variables constant. This conditional interpretation is useful, but it is sensitive to the set of correlated predictors and their scaling. A coefficient should not be read as a causal effect unless the design supports that claim. In a predictive PD model, its principal role is to make the contribution of a defined feature explicit.

Origination and behavioural variables serve different purposes. Origination characteristics measure the risk profile at the start of the loan. Monthly balance, interest rate, age, remaining term, and delinquency history update that profile as the account evolves. Behavioural variables can dominate short-horizon prediction because they record deterioration that has already begun. Their construction must stop at the prediction month. A rolling count that incorporates month $t + 1$, even indirectly, would contaminate the information set used for a month- t forecast.

Repeated loan-month observations create another distinction from a cross-sectional score-

card. The same loan contributes several observations, and its static characteristics recur while its age, balance, payment record, and common macroeconomic environment change. The resulting sample can be large without containing an equally large number of independent economic periods. Chronological validation is therefore more informative for temporal transport than a random division of loan-month rows. Random allocation would place observations from the same calendar month, and often the same loan, on both sides of the evaluation boundary.

A borrower-only logistic model is static in calendar time unless temporal terms are introduced explicitly. Its coefficients imply that a unit change in a borrower or loan characteristic has the same log-odds effect in all months. This is a strong restriction for mortgage PD. The risk associated with leverage, interest cost, or recent delinquency may differ when labour income, financing conditions, and collateral values deteriorate together. A static scorecard can retain useful cross-sectional discrimination while its probability levels drift as the economic environment changes.

Static coefficients can fail in two related ways. The intercept may no longer represent the prevailing base rate, and the slopes may no longer represent the prevailing relationship between loan characteristics and the outcome. Recalibrating the intercept can address a broad level shift. It cannot correct a change in the relative importance of credit score, leverage, interest cost, or payment history. A model that allows only an intercept adjustment remains static in its borrower sensitivities.

The distinction matters for evaluation. If every predicted probability is too high by a similar amount, the model can preserve ROC-AUC while producing poor Brier score and log loss. If the slopes change, ranking can also deteriorate. A complete assessment should therefore compare probability scores, calibration levels, and discrimination. A single measure cannot identify whether error arises from a base-rate shift, weak separation, or both.

Macroeconomic credit models address part of this problem by adding observed economic variables to the loan-level equation. Unemployment, interest rates, inflation, house prices,

and related variables connect default forecasts to common conditions. Evidence from survival and stress-testing models shows that macroeconomic variables can improve the temporal responsiveness of retail credit models (Bellotti and Crook, 2009, 2013). The usual additive form shifts the log-odds by the same macroeconomic contribution for all loans. It does not imply that a loan characteristic becomes more or less consequential as the economy moves into stress. A large correlated macro vector can also weaken stability and interpretation.

An additive macroeconomic model has the form $\text{logit}(\pi_{i,t}) = \beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma$. The same $M_t^\top \gamma$ is applied to every loan observed in month t . Borrowers can still receive different PDs because their $X_{i,t}$ values differ, but the marginal log-odds effect of each borrower feature remains β_j in every month. The macroeconomic vector changes the common level of risk rather than the cross-sectional sensitivity to borrower information.

This form is useful as a baseline because it tests whether direct macroeconomic measurements add information after loan-level variables are included. It is also demanding to estimate when several indicators describe related economic mechanisms. Policy rates, mortgage rates, Treasury yields, and spreads are correlated. Unemployment, claims, payrolls, income, and sentiment overlap in their description of labour and household conditions. Regularisation can stabilise their coefficients, but the fitted vector can remain difficult to summarise economically.

The timing of macroeconomic variables must match the information set. Levels, changes, spreads, and growth rates use different amounts of historical information. A twelve-month growth rate at month t is available from observations at t and $t - 12$. A monthly change is available from t and $t - 1$. These backward-looking transformations are consistent with a month- t forecast. Revisions to public macroeconomic series create a separate real-time issue when current-vintage data are used to describe earlier months. The present study does not reconstruct historical data vintages, so its macroeconomic inputs should be interpreted as revised series aligned by calendar month.

Hidden Markov models provide a parsimonious representation of persistent latent condi-

tions. The observed macroeconomic vector is assumed to arise from one of a finite number of unobserved states, while the state follows a first-order Markov chain. State persistence distinguishes an HMM from a contemporaneous classification rule. The model can summarise several indicators as a probability distribution over economic states rather than assigning an observed month to a regime without uncertainty (Rabiner, 1989; Hamilton, 1989).

A two-state HMM separates the macroeconomic sequence into two fitted distributions. The state labels are statistical and have no inherent economic meaning. One state becomes Stress only after its fitted emission profile is compared with the orientation of the inputs. This post-estimation orientation avoids assigning meaning to the arbitrary numeric state labels. It also makes the definition reproducible: the stress state is the state with the larger combined mean across variables that have been oriented towards adverse conditions.

The transition matrix represents persistence. Large diagonal probabilities imply that the latent state tends to remain unchanged from one month to the next. This persistence can reduce month-to-month noise in the regime signal. It can also produce probabilities close to zero or one when the fitted state distributions are well separated. Such certainty is not automatically evidence of a superior economic signal. Near-degenerate probabilities may leave too little variation for interaction effects to be distinguished in a later test period.

The Gaussian emission assumption is a modelling restriction. Within each state, the factor vector is represented by a multivariate normal distribution with a state-specific mean and covariance matrix. Full covariance matrices allow the relationships among factors to differ by state. They require more parameters than diagonal covariance matrices and can be sensitive when the number of monthly observations is limited. Reducing the macroeconomic inputs to a small number of interpretable factors limits this parameter burden without claiming that the factors exhaust every dimension of the macroeconomy.

The distinction between filtered and smoothed state probabilities is essential in forecasting. A smoothed probability conditions on the complete observed sequence and therefore uses information after month t . A filtered probability conditions on observations through

month t . The filtered stress probability, denoted $q_t = \Pr(R_t = \text{Stress} \mid Z_{1:t}; \hat{\Theta})$, is the appropriate state output for a month- t PD equation. It remains conditional on the method used to estimate $\hat{\Theta}$. If HMM parameters and transformations are fitted over the full sample, forward filtering prevents direct use of future observations in the recursion but does not create a fully real-time estimate.

This distinction produces two separate information questions. The first concerns state inference given fixed parameters. Forward filtering answers that question using $Z_{1:t}$. The second concerns parameter estimation. A fully real-time design would estimate transformations, scaling parameters, and HMM parameters using information available at each forecast origin. Using a full-window estimate answers a narrower question: what filtered state path is implied by parameters fitted to the complete study window? The resulting signal is more defensible than a smoothed path for downstream prediction, but it should not be described as strictly out of sample.

The continuous probability q_t contains more information than the hard indicator $\mathbb{I}(q_t \geq 0.5)$. Values near 0.5 express state uncertainty. Values near zero or one express stronger assignment under the fitted model. A hard threshold discards this gradation and can create an artificial discontinuity. It is still useful for descriptive counts and diagnostic slices. It is unsuitable for a normal-versus-stress comparison when every observation in an evaluation period lies on the same side of the threshold.

Regime-dependent credit models allow default processes to vary across persistent economic states. Existing applications have examined cyclical rating migration and hidden-state default intensity (Bangia et al., 2002; Yu et al., 2018). This literature establishes that latent regimes can enter credit-risk models, but it does not settle whether an HMM improves a loan-level mortgage PD after borrower and payment-history information is included. It also leaves a choice between a hard regime indicator and a continuous state probability. A continuous probability preserves uncertainty and avoids an abrupt change at an arbitrary threshold.

Portfolio-level regime models and loan-level regime-aware models answer different questions. A portfolio model can allow a common transition matrix or default intensity to change with the state. A loan-level model must also represent cross-sectional heterogeneity among loans observed in the same month. The common state alone cannot distinguish two borrowers. Borrower information remains the primary source of cross-sectional variation, while the regime signal supplies a shared time-varying context.

This layered interpretation places the HMM between the macroeconomic data and the PD equation. It is a dimension-reduction and state-inference component rather than a replacement for the credit model. The HMM is estimated from monthly macroeconomic factors. The logistic model is estimated from loan-month outcomes. Their parameters have different meanings and are supported by different effective sample sizes. Millions of loan-months do not eliminate uncertainty in a regime process estimated from slightly more than a decade of monthly observations.

A regime-aware loan-level logistic model can include q_t as a level effect and interact it with selected components of $X_{i,t}$. The log-odds contribution of feature j then becomes $\beta_j + q_t\theta_j$. This structure is more flexible than an additive macroeconomic model and remains interpretable by construction. Interpretation still requires care. The sign and magnitude of a standardised interaction coefficient depend on the included main effects, feature scaling, and correlation among variables. The structure identifies which sensitivities are allowed to vary; it does not by itself establish a causal effect of the macroeconomic state.

The level term and interaction terms serve distinct purposes. The coefficient on q_t shifts the log-odds for every loan after accounting for $X_{i,t}$. Each interaction changes the coefficient on one selected loan-level feature as q_t changes. A stress-level-only model can therefore be compared with the interaction model. If the level-only model improves but the interaction model does not, the evidence favours a common macroeconomic shift. If the interaction model improves while the level-only model does not, the selected changing sensitivities are the more plausible source of the predictive difference.

Interaction selection should remain limited when interpretation is a central aim. Interacting every loan-level variable with every macroeconomic series would create a large model whose coefficients are difficult to separate. A smaller set based on credit quality, leverage, interest cost, and recent payment performance preserves the economic question. Regularisation remains necessary because the main effects and their products with a common q_t can be strongly correlated, especially when q_t spends long periods near an endpoint.

When q_t has little variation in the test period, the regime-aware model can still generate different predictions because the interaction coefficients were estimated using earlier variation and because borrower features vary across loans. Its held-out performance can therefore be measured. What cannot be established is performance across distinct test regimes. A favourable pooled score during an all-stress period is evidence about that terminal period, not evidence of a stable normal-to-stress contrast.

Interpretability is relevant because a higher-capacity specification can improve fit while obscuring the source of a probability change. A regularised logistic model retains an explicit functional form and permits direct inspection of borrower–regime terms. This advantage does not exempt the model from empirical validation. A transparent model can still be unstable, poorly calibrated, or sensitive to the estimation window. Interpretability concerns the form of the model. Predictive validity concerns its behaviour on data not used for estimation.

Regularisation changes the interpretation of fitted coefficients. An L_2 penalty shrinks correlated coefficients towards zero but does not set most of them exactly to zero. The resulting model can be more stable than an unpenalised fit, particularly with macroeconomic and interaction terms. Coefficients remain conditional associations on the standardised scale. Their presence documents the modelled pathway from a predictor to log-odds; their magnitude should not be presented as an unpenalised structural parameter.

Interpretability also depends on reporting the complete specification. A label such as “regime-aware” is insufficient unless the main effects, regime input, and interaction set are defined. The calibration map is part of the final probability model because it changes

probability levels after the logistic fit. A transparent account should therefore distinguish the raw logistic probability, the calibration transformation estimated on validation data, and the final probability used for evaluation.

Calibration and discrimination answer different questions. ROC-AUC measures the ordering of cases and controls, while PR-AUC focuses more directly on positive-class retrieval when defaults are rare (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Neither measure assesses whether predicted probabilities match observed frequencies. The Brier score and log loss are proper scoring rules that assess probability forecasts (Brier, 1950; Gneiting and Raftery, 2007). Calibration curves compare predicted and observed rates across probability groups. A Brier decomposition separates reliability, resolution, and outcome uncertainty, although its numerical values depend on the grouping rule.

The Brier score averages squared probability errors. It is sensitive to calibration and discrimination but is often dominated by the large number of non-events when the outcome is rare. Log loss gives a larger penalty to confident errors because it is logarithmic. The two scores can rank models differently. Agreement between them strengthens a comparison, while disagreement requires a metric-specific interpretation. Neither score should be converted into a claim of economic significance without considering the size of the difference and the dependence among repeated observations.

ROC-AUC is invariant to monotonic recalibration. A model can retain the same ROC-AUC after its probabilities are shifted substantially. PR-AUC is also primarily a ranking measure, but its scale depends on event prevalence. Reporting the event rate makes its magnitude interpretable. In a rare-event setting, PR-AUC can distinguish models whose ROC curves appear similar because it places more emphasis on the precision of high-ranked positive predictions.

Calibration has several levels. Mean calibration compares the average predicted PD with the overall event rate. Calibration-in-the-large can be poor even when a curve has acceptable local shape. A calibration slope assesses whether probabilities are too dispersed

or too compressed. Grouped curves show how predicted and observed rates compare across selected bins. These summaries complement one another. A model can have a lower Brier score than a comparator while both models materially overpredict the mean event rate.

The Murphy decomposition expresses a grouped Brier score as reliability minus resolution plus uncertainty. Reliability is smaller when average predictions within bins are close to observed rates. Resolution is larger when observed rates differ across bins. Uncertainty depends only on the overall event rate. Because the decomposition uses bins, it approximates rather than reproduces the ungrouped Brier score. Its components should be used to explain a result, not to replace the primary ungrouped score.

Post-estimation calibration can correct systematic distortion in raw probabilities. Platt scaling estimates an intercept and slope on the logit of the raw probability. Intercept-only recalibration adjusts the level while preserving the slope. Isotonic regression estimates a non-decreasing mapping with greater flexibility. These methods should be fitted without using the test outcomes. Chronological validation is therefore important for both model selection and calibration. It also reveals whether a result is stable across time rather than a consequence of random mixing across economic periods.

Platt scaling can correct both a broad level shift and excessive or insufficient dispersion on the logit scale. Intercept-only recalibration is more restrictive and leaves ranking and spread unchanged. Isotonic regression makes fewer shape assumptions but can adapt closely to the validation sample. Comparing these mappings on the same held-out period provides a sensitivity analysis. Choosing a new primary method after inspecting test outcomes would weaken the separation between validation and evaluation.

Temporal validation carries information beyond a single test score. A model that is best on validation and worse on test may have deteriorated as the population or economy changed. A model that is worse on validation and better on test may be suited to the terminal period without demonstrating stable superiority. Both patterns should be reported. The final interpretation should distinguish a held-out result from a recurring result across evaluation

windows.

Portfolio calibration adds another level of assessment. Summing loan-level PDs within a month gives an expected number of defaults under the model. Comparing that quantity with the observed monthly count tests whether pooled probability performance translates into useful timing and level estimates. A model can improve the average Brier score by reallocating risk across loans while still overstating the total number of defaults. This is not a contradiction. It reflects different aggregations of the same probability forecasts.

The evidence supports a focused empirical comparison. Borrower-only logistic regression provides the loan-level benchmark. An additive macroeconomic logit tests whether observed macro variables shift PD levels. A regime-aware logit tests whether a filtered stress probability and selected borrower–regime interactions improve probability forecasts. The comparison should place calibration and proper scores alongside discrimination. It should also examine monthly predicted-default counts because a model can improve pooled loan-level scores while misstating the timing or level of portfolio risk.

This comparison gives each model a defined role. The borrower-only model establishes the value of cross-sectional and behavioural information without common economic inputs. The macro-augmented model tests a direct additive treatment of observed conditions. The regime-aware model tests a compressed latent-state signal and changing sensitivities. Diagnostic variants can separate the stress level from its interactions and assess whether the full macro vector contributes beyond the regime probability. The empirical conclusion should follow the joint pattern of these comparisons rather than the label attached to any one specification.

Methodology

3.1 Data and Study Population

The study combined Freddie Mac Single-Family Loan-Level Dataset sample files with macroeconomic series from Federal Reserve Economic Data (FRED) (Freddie Mac, 2026; Federal Reserve Bank of St. Louis, 2026). Freddie Mac origination records supplied initial borrower and contract characteristics. Monthly performance records supplied delinquency status, balance, interest rate, loan age, remaining term, and termination information. The available sample cohorts covered 2015–2025. The twelve-month follow-up rule left eligible observations from the 2015–2024 cohorts and a common endpoint of September 2024.

Table 3.1: Data components and roles in the retained specification

Component	Information	Empirical role
Freddie Mac origination records	Credit score, leverage, balance, rate, term, insurance, units, and borrower count	Initial borrower and contract risk characteristics
Freddie Mac monthly performance records	Delinquency status, balance, current rate, loan age, remaining term, and termination	Risk set, twelve-month outcome, and time-varying loan features
FRED macroeconomic series	Prices, rates, labour indicators, income, house prices, financial conditions, and sentiment	Transformed macro baseline and four HMM factors
Filtered HMM output	Monthly probability of the stress state	Direct regime term and borrower–regime interactions

Origination and performance files were linked by the loan sequence identifier. Calendar

dates were normalised to the first day of each month. Delinquency status in the servicing data was converted from months delinquent to days past due by multiplying non-negative numeric values by 30. Loan-level records were ordered by identifier and month before the outcome and payment-history variables were calculated.

The unit of analysis was a loan-month. Let $D_{i,t}$ denote days past due for loan i in month t . A loan-month entered the risk set when the loan was active, $D_{i,t} < 90$, the servicing record extended through $t + 12$, and no terminal servicing event occurred on or before $t + 12$. The eligible set was

$$\mathcal{R} = \{(i, t) : D_{i,t} < 90, t + 12 \leq T_i^{\max}, T_i^{\text{term}} > t + 12 \text{ or } T_i^{\text{term}} \text{ is missing}\}, \quad (3.1)$$

where $T_i^{\max}$ is the last observed month and T_i^{term} is the first terminal month. This rule excluded observations with incomplete forward follow-up instead of treating them as non-events.

The primary outcome was

$$Y_{i,t}^{(12)} = \mathbb{I}\left\{\max_{h=1,\dots,12} D_{i,t+h} \geq 90\right\}, \quad (i, t) \in \mathcal{R}. \quad (3.2)$$

The conditional PD was $\pi_{i,t}^{(12)} = \Pr(Y_{i,t}^{(12)} = 1 \mid \mathcal{F}_t)$. All predictors were measured at origination or constructed from information observed by month t .

3.2 Loan-Level Predictors

The borrower and loan vector $X_{i,t}$ contained 14 origination or current-account variables and 18 payment-history variables. The first group comprised credit score, mortgage-insurance percentage, number of units, original combined loan-to-value ratio, original debt-to-income ratio, original unpaid principal balance, original loan-to-value ratio, original interest rate, original term, number of borrowers, loan age, remaining months to legal maturity, current

unpaid principal balance, and current interest rate.

The payment-history group comprised current delinquency in months; current 30- and 60-day delinquency indicators; recent 30- and 60-day delinquency within three months; ever-30- and ever-60-day indicators; months since 30- and 60-day delinquency; counts of 30- and 60-day delinquent months over three, six, and twelve months; a cure-from-30-days indicator; cumulative 30-day cures; and months since a 30-day cure. These features were calculated within loan and used observations through month t .

Table 3.2: Loan-level predictor groups

Group	Variables
Origination and account	Credit score; mortgage-insurance percentage; number of units; original combined and ordinary loan-to-value ratios; original debt-to-income ratio; original and current unpaid principal balance; original and current interest rate; original term; number of borrowers; loan age; remaining months to legal maturity
Current delinquency	Delinquency months; current 30-day indicator; current 60-day indicator
Recent and cumulative history	Recent 30- and 60-day indicators; ever-30- and ever-60-day indicators; months since 30- and 60-day delinquency
Rolling history	Counts of 30- and 60-day delinquent months over three, six, and twelve months
Cure history	Cure from 30 days past due; cumulative 30-day cures; months since a 30-day cure

The rolling counts were based on cumulative within-loan event counts and their lags. A recent indicator equalled one when the corresponding three-month count was positive. A cure occurred when a loan moved from at least 30 days past due in month $t - 1$ to below 30 days past due in month t . For loans with no prior delinquency or cure, the elapsed-month variable was measured from the first observed loan month plus one. This convention provided a finite history measure without using a later event date.

3.3 Macroeconomic Variables and Factors

The macroeconomic data contained CPI, the Federal Funds rate, unemployment, the FHFA house-price index, the 30-year mortgage rate, the 10-year Treasury rate, the 10-year minus 2-year Treasury spread, initial unemployment claims, non-farm payrolls, real disposable personal income, the National Financial Conditions Index, and consumer sentiment. Weekly and daily observations were averaged to monthly frequency. The retained transformations were twelve-month inflation, the policy rate, unemployment and its monthly change, twelve-month house-price growth, the mortgage rate, the mortgage–Treasury spread, the long rate and its monthly change, the yield curve, initial claims and their monthly change, twelve-month payroll and real-income growth, financial conditions, consumer sentiment, and its monthly change.

The HMM did not receive the 17 transformed series directly. Each series was oriented so that a larger value represented greater stress and standardised. Available series were then averaged within four groups: housing/collateral, rate/refinancing, labour/income, and financial conditions. For group g ,

$$F_{t,g} = \frac{1}{|\mathcal{G}_g|} \sum_{j \in \mathcal{G}_g} \frac{a_j M_{t,j} - \overline{a_j M_j}}{s(a_j M_j)}, \quad (3.3)$$

where $a_j \in \{-1, 1\}$ sets the adverse direction. The four factors were standardised again before HMM estimation. Equal weighting retained a direct mapping from the source series to each factor.

The HMM estimation window was January 2014–September 2024. It included a twelve-month margin before the first mortgage observation. The factor transformations, scaler, and HMM parameters were estimated once over this complete window.

Inflation remained in the 17-variable observed macroeconomic vector used by the additive baseline. It was not assigned to one of the four retained HMM factor groups. The HMM source set therefore differed from the complete macro-augmented logistic input. The additive

Table 3.3: Macroeconomic transformations and HMM grouping

Factor	Source transformations	Adverse orientation
Housing/collateral	Twelve-month house-price growth	Lower growth
Rate/refinancing	Policy rate; mortgage rate; mortgage–Treasury spread; 10-year rate and monthly change; 10-year minus 2-year spread	Higher rates and spread; lower yield curve
Labour/income	Unemployment and change; claims and change; twelve-month payroll and real-income growth; sentiment and change	Higher unemployment and claims; lower growth and sentiment
Financial conditions	National Financial Conditions Index	Higher index

baseline received every available transformed macro series, while the HMM received the four grouped factors recorded in the retained HMM artefact.

3.4 Two-State Hidden Markov Model

Let $R_t \in \{0, 1\}$ denote the latent macroeconomic state and let Z_t be the four-dimensional standardised factor vector. The state process followed a first-order Markov chain,

$$\Pr(R_t = k \mid R_{t-1} = j) = p_{jk}, \quad P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}, \quad (3.4)$$

and the emissions followed state-specific multivariate Gaussian distributions,

$$Z_t \mid R_t = k \sim \mathcal{N}(\mu_k, \Sigma_k), \quad k \in \{0, 1\}. \quad (3.5)$$

Both covariance matrices were unrestricted. The model used two states, full covariance matrices, a maximum of 500 estimation iterations, and a fixed random seed of 42. Parameters were estimated by the expectation-maximisation procedure for Gaussian HMMs (Baum et al., 1970; Bilmes, 1998).

For a latent state path $R_{1:T}$ and observed factors $Z_{1:T}$, the joint density was

$$f(Z_{1:T}, R_{1:T}; \Theta) = \delta_{R_1} \prod_{t=2}^T p_{R_{t-1}, R_t} \prod_{t=1}^T \phi_4(Z_t; \mu_{R_t}, \Sigma_{R_t}), \quad (3.6)$$

where δ is the initial state distribution and ϕ_4 is the four-dimensional Gaussian density. The expectation step calculated posterior state and transition probabilities under the current parameter values. The maximisation step updated the initial probabilities, transition matrix, state means, and covariance matrices.

The fitted state labels were oriented after estimation. If $\hat{\mu}_{k,g}$ is the fitted mean of adverse-oriented factor g in state k , the state score was

$$s_k = \sum_{g=1}^4 \hat{\mu}_{k,g}. \quad (3.7)$$

The state with the larger s_k was labelled Stress. This rule used all four HMM inputs and did not define stress from the frequency or chronology of the fitted state.

The state with the larger sum of adverse-oriented fitted factor means was labelled Stress. The forward recursion produced

$$\alpha_1(k) \propto \delta_k f_k(Z_1), \quad (3.8)$$

$$\alpha_t(k) \propto f_k(Z_t) \sum_{j=0}^1 \alpha_{t-1}(j) p_{jk}, \quad (3.9)$$

with normalisation at each month. The retained macroeconomic input was

$$q_t = \Pr(R_t = \text{Stress} \mid Z_{1:t}; \hat{\Theta}). \quad (3.10)$$

This was a filtered rather than smoothed probability. It did not condition on factor observations after month t in the state recursion. It remained conditional on transformations and HMM parameters estimated over the full January 2014–September 2024 window.

The fitted output also contained $\mathbb{I}(q_t \geq 0.5)$, three-, six-, and twelve-month lags of q_t , the monthly change Δq_t , and the duration of the current sequence above 0.5. The primary regime-aware model used only q_t . The hard indicator was used for descriptive regime slices. The smaller dynamic diagnostic used q_t , Δq_t , and duration. The lagged values were retained in the HMM output but were not included in the reported dynamic model.

3.5 Loan-Level PD Models

All PD specifications were regularised logistic models. Numerical inputs and interaction terms were standardised within each fitted pipeline. The borrower-only baseline was

$$\pi_{i,t}^{(B)} = \sigma(\beta_0 + X_{i,t}^\top \beta), \quad (3.11)$$

where $\sigma(u) = (1 + e^{-u})^{-1}$. The macro-augmented baseline added the 17 transformed macroeconomic variables M_t :

$$\pi_{i,t}^{(M)} = \sigma(\beta_0 + X_{i,t}^\top \beta + M_t^\top \gamma). \quad (3.12)$$

The regime-aware model used q_t and selected interactions:

$$\pi_{i,t}^{(R)} = \sigma(\beta_0 + X_{i,t}^\top \beta + \gamma_R q_t + q_t W_{i,t}^\top \theta), \quad (3.13)$$

where $W_{i,t}$ contained credit score, original combined loan-to-value ratio, original debt-to-income ratio, original loan-to-value ratio, original interest rate, current delinquency in months, recent 60-day delinquency within three months, and months since 30-day delinquency. The marginal log-odds coefficient for an interacted variable $W_{i,t,j}$ was $\beta_j + q_t \theta_j$. The specification therefore allowed selected loan-level sensitivities to vary continuously with the estimated stress probability.

Three additional models separated components of this structure. A stress-level model added q_t to $X_{i,t}$ without interactions. A regime-aware plus raw-macro model combined

Table 3.4: Model hierarchy

Model	Inputs
Borrower-only	$X_{i,t}$
Macro-augmented	$X_{i,t}$ and the 17 transformed macroeconomic variables M_t
Stress level only	$X_{i,t}$ and q_t
Regime-aware	$X_{i,t}$, q_t , and $q_t W_{i,t}$
Regime-aware plus macro	$X_{i,t}$, M_t , q_t , and $q_t W_{i,t}$
Dynamic stress	$X_{i,t}$, q_t , $q_t W_{i,t}$, Δq_t , and stress duration

Equation (3.13) with M_t . A dynamic-stress diagnostic added $\Delta q_t = q_t - q_{t-1}$ and the number of consecutive months for which $q_t \geq 0.5$. These models were evaluated on the same target and chronological samples.

The implementation used stochastic-gradient logistic regression with an L_2 penalty. Candidate penalty values were $\{3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}\}$. Each value was evaluated on deterministic samples of 500,000 training and 500,000 validation observations. The value with the lowest validation Brier score was selected, and the model was then fitted to the complete training sample.

For coefficient vector ω and intercept β_0 , estimation minimised a penalised Bernoulli objective of the form

$$- \sum_{(i,t) \in \mathcal{R}_{\text{train}}} \left[Y_{i,t}^{(12)} \log \pi_{i,t} + (1 - Y_{i,t}^{(12)}) \log(1 - \pi_{i,t}) \right] + \lambda \|\omega\|_2^2. \quad (3.14)$$

The intercept was separate from the penalised feature vector. Standardisation was fitted within the model pipeline. The coefficient artefact therefore recorded effects on the standardised scale. The empirical analysis used the coefficients to define an explicit predictive model rather than to make causal claims.

3.6 Chronological Validation and Calibration

The 117 model months were ordered before splitting. The first 70% formed the training period, the next 15% formed the validation period, and the final 15% formed the test period. This produced January 2015–September 2021 for training, October 2021–March 2023 for validation, and April 2023–September 2024 for testing. Model selection used the validation period. Test outcomes were reserved for final evaluation.

Raw probabilities were calibrated using the complete validation predictions. The primary method was Platt scaling,

$$\hat{\pi}^{\text{Platt}} = \sigma(a + b \logit(\hat{\pi}^{\text{raw}})). \quad (3.15)$$

Intercept-only recalibration and isotonic regression were retained as sensitivity analyses. The fitted calibration maps were applied unchanged to training, validation, and test predictions. All primary comparisons used the Platt-calibrated probability.

The intercept-only map had the form

$$\hat{\pi}^{\text{Int}} = \sigma(c + \logit(\hat{\pi}^{\text{raw}})), \quad (3.16)$$

where c was selected so that the validation mean matched the validation event rate. Isotonic regression estimated a non-decreasing mapping from raw validation probabilities to validation outcomes, with values outside the fitted range clipped to the fitted boundaries. Neither sensitivity map used test outcomes during fitting.

3.7 Evaluation Measures

The primary probability measures were the Brier score and log loss. For an evaluation sample $\mathcal{S}$,

$$BS = \frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} (\hat{\pi}_{i,t} - Y_{i,t}^{(12)})^2, \quad (3.17)$$

$$LL = -\frac{1}{|\mathcal{S}|} \sum_{(i,t) \in \mathcal{S}} \left[Y_{i,t}^{(12)} \log \hat{\pi}_{i,t} + (1 - Y_{i,t}^{(12)}) \log(1 - \hat{\pi}_{i,t}) \right]. \quad (3.18)$$

Lower values indicate better probability forecasts. The Brier score was also decomposed over ten equal-count probability bins into reliability, resolution, and uncertainty (Brier, 1950; Murphy, 1973; Dimitriadis et al., 2020).

For bin b , let n_b be its size, $\bar{\pi}_b$ its mean prediction, $\bar{y}_b$ its observed event rate, and $\bar{y}$ the overall event rate. The grouped components were

$$\text{Reliability} = \sum_{b=1}^{10} \frac{n_b}{n} (\bar{\pi}_b - \bar{y})^2, \quad (3.19)$$

$$\text{Resolution} = \sum_{b=1}^{10} \frac{n_b}{n} (\bar{y}_b - \bar{y})^2, \quad (3.20)$$

$$\text{Uncertainty} = \bar{y}(1 - \bar{y}). \quad (3.21)$$

The grouped Brier diagnostic equalled reliability minus resolution plus uncertainty. Equal-count bins were constructed by ranking predictions before division into deciles. The grouped value was not expected to equal the ungrouped score exactly.

ROC-AUC and PR-AUC measured discrimination. PR-AUC was reported because the serious-delinquency outcome was rare (Davis and Goadrich, 2006; Saito and Rehmsmeier, 2015). Calibration curves compared mean predicted PD with observed event frequency across ten bins. Annual test-period Brier scores described temporal variation. The monthly portfolio diagnostic compared observed defaults, $D_t = \sum_i Y_{i,t}^{(12)}$, with predicted defaults,

$\hat{D}_t = \sum_i \hat{\pi}_{i,t}$, using mean absolute count error and mean absolute relative error.

Test observations were also divided into ten quantile bins of q_t . Each bin recorded its stress-probability range, event rate, mean prediction, and model metrics. These bins described gradients within the terminal period. They did not supply independent macroeconomic episodes because each observation in every bin had a hard-stress indicator equal to one.

No rolling-origin HMM re-estimation, block bootstrap, loan-clustered uncertainty interval, or alternative HMM specification was performed. The analysis therefore reports point estimates for one chronological split. It does not treat the number of loan-months as the number of independent macroeconomic observations.

Results

4.1 Sample and Outcome

The eligible panel contained 15,461,471 loan-months from 438,713 loans across the 2015–2024 cohorts. It contained 167,056 twelve-month transitions to 90 or more days past due, an event rate of 1.0805%. The training sample contained 8,179,648 observations with an event rate of 1.5191%. The validation sample contained 3,175,548 observations with an event rate of 0.4982%. The test sample contained 4,106,275 observations and 26,983 events, an event rate of 0.6571%. The variation in event rates across the chronological samples shows that the probability level changed materially over time.

Table 4.1 reports the retained cohort coverage. Earlier cohorts contributed longer histories and more loan-months. The 2023 and 2024 cohorts contributed fewer observations because the model-ready data ended in September 2024 and each included loan-month required twelve subsequent observed months. Cohort event rates were not monotonic. They ranged from 0.4361% for the 2020 cohort to 2.0893% for the 2019 cohort.

The cohort table describes origination groups, while the training, validation, and test samples were divided by observation month. A loan from an early cohort could therefore contribute to more than one chronological split as it aged. No future outcome for a test-period loan-month entered model selection or probability calibration. The lower validation event rate and later increase in the test rate created a direct test of temporal calibration transfer.

Table 4.1: Eligible sample by origination cohort

Cohort	Loans	Loan-months	Events	Event rate
2015	46,574	2,756,432	21,296	0.007726
2016	47,303	2,653,282	24,896	0.009383
2017	47,144	2,077,915	30,191	0.014529
2018	44,005	1,360,031	28,405	0.020886
2019	39,514	1,229,576	25,689	0.020893
2020	42,751	1,744,236	7,606	0.004361
2021	47,277	1,691,649	8,233	0.004867
2022	47,807	1,187,079	13,177	0.011100
2023	46,158	630,299	6,653	0.010555
2024	30,180	130,972	910	0.006948
Total	438,713	15,461,471	167,056	0.010805

4.2 Estimated Macroeconomic Regimes

The HMM used 129 monthly factor observations from January 2014 to September 2024. The fitted stress probability ranged from approximately 1.63×10^{-102} to a value numerically equal to one over that window. Ninety-nine months were below the hard-state threshold of 0.5 and 30 months were above it. Figure 4.1 shows the filtered probability path.

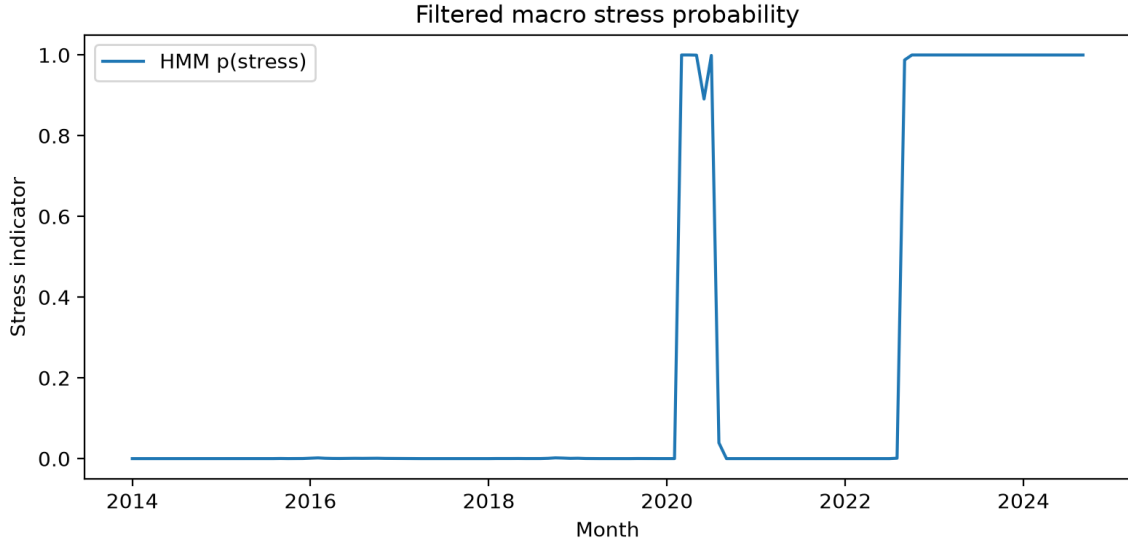


Figure 4.1: Forward-filtered probability of the HMM stress state, January 2014–September 2024. Factor transformations and HMM parameters were estimated over the full displayed window.

The probability was nearly saturated during the test period. It ranged from 0.999903 to a value numerically indistinguishable from one, and every test observation had $q_t \geq 0.5$. The test data therefore did not contain a hard normal-regime comparison. Quantile bins separated only small numerical differences near one. The retained HMM answered the first research question by assigning a persistent stress state throughout the terminal test period, but it provided weak variation for estimating cross-regime test performance.

4.3 Validation Results

The borrower-only model performed best on the validation sample under the primary Platt calibration. Its Brier score was 0.004645 and its log loss was 0.026122. The regime-aware values were 0.004682 and 0.027021. The macro-augmented values were 0.004683 and 0.026850. Borrower-only also had the highest validation ROC-AUC, 0.798634, and PR-AUC, 0.114809. The corresponding regime-aware values were 0.767182 and 0.105560. The validation evidence did not show an advantage for the regime-aware specification.

Table 4.2: Validation performance using Platt-calibrated probabilities

Model	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.004645	0.026122	0.798634	0.114809
Macro-augmented	0.004683	0.026850	0.775172	0.106811
Stress level only	0.004685	0.026627	0.786294	0.105784
Regime-aware	0.004682	0.027021	0.767182	0.105560
Regime-aware plus macro	0.004722	0.027404	0.756329	0.098194
Dynamic stress	0.004694	0.026656	0.784194	0.098966

The stress-level-only model had lower validation log loss than the regime-aware model but a slightly higher Brier score. The regime-aware plus macro model was weakest on validation Brier and log loss. The dynamic-stress model did not improve on borrower-only. The ranking therefore did not support adding complexity on the basis of the validation period.

Isotonic calibration reduced the validation Brier score to 0.004629 for borrower-only and 0.004647 for regime-aware. These values did not change the validation ordering. Platt scaling

remained the primary calibration method specified in the implementation.

The regularisation choices also differed. Borrower-only, macro-augmented, stress-level-only, regime-aware plus macro, and dynamic stress selected 3×10^{-5} . Regime-aware selected 3×10^{-6} . These values were selected from the same candidate set using deterministic training and validation samples. The later test ranking was not used to choose them.

4.4 Held-Out Test Results

Table 4.3 reports the primary Platt-calibrated test results. The regime-aware model had the lowest Brier score and log loss and the highest ROC-AUC and PR-AUC. Relative to borrower-only, its Brier score fell from 0.006055 to 0.006015, a reduction of 0.66%. Its log loss fell from 0.033106 to 0.032945, a reduction of 0.48%. ROC-AUC increased by 0.005276 and PR-AUC increased by 0.011736.

Table 4.3: Held-out test performance using Platt-calibrated probabilities

Model	Mean PD	Brier	Log loss	ROC-AUC	PR-AUC
Borrower-only	0.010747	0.006055	0.033106	0.821979	0.148361
Macro-augmented	0.009743	0.006078	0.033453	0.805984	0.133367
Stress level only	0.013348	0.006089	0.034349	0.825747	0.150199
Regime-aware	0.010802	0.006015	0.032945	0.827256	0.160097
Regime-aware plus macro	0.011313	0.006105	0.034187	0.804293	0.131448
Dynamic stress	0.000687	0.006472	0.045723	0.745641	0.077918

Note: $n = 4,106,275$ loan-months; 26,983 events; event rate 0.006571. Lower Brier score and log loss are better. Higher ROC-AUC and PR-AUC are better.

The nested specifications qualified this result. Adding q_t without interactions produced a Brier score of 0.006089 and log loss of 0.034349. Adding the full macro vector to the regime-aware model produced 0.006105 and 0.034187. The dynamic-stress model substantially underpredicted the test event rate and performed worst on all four reported measures. The held-out advantage was specific to the selected regime-aware interaction structure.

The test ranking differed from the validation ranking. This limits the strength of the conclusion. The evidence supports a modest terminal-period improvement in pooled loan-

level performance. It does not show that the improvement was stable across the validation and test periods.

The absolute test differences were small relative to the scale of the scores. The Brier difference between regime-aware and borrower-only was 0.0000397. The log-loss difference was 0.0001603. The report tables do not contain uncertainty intervals for these paired differences. The observed ordering is exact for the retained predictions, but its sampling uncertainty was not estimated.

4.5 Calibration

The test event rate was 0.006571. The mean Platt-calibrated PD was 0.010747 for borrower-only, 0.009743 for macro-augmented, and 0.010802 for regime-aware. Each core model over-predicted the mean event rate. Figure 4.2 shows that validation-based calibration did not remove this temporal shift.

The ten-bin Brier decomposition gave reliability of 0.0000293 and resolution of 0.0001017 for the regime-aware model. Borrower-only had reliability of 0.0000549 and resolution of 0.0001006. The regime-aware grouped diagnostic therefore had lower reliability error and similar resolution. The macro-augmented model had lower grouped reliability, 0.0000247, but worse ungrouped Brier score and discrimination. The grouped decomposition did not reproduce the complete model ordering.

Table 4.4 compares calibration maps for the two principal models. Isotonic calibration slightly improved the regime-aware Brier score and log loss relative to Platt. It worsened both measures for borrower-only. This was a sensitivity result because the primary probability column used Platt scaling.

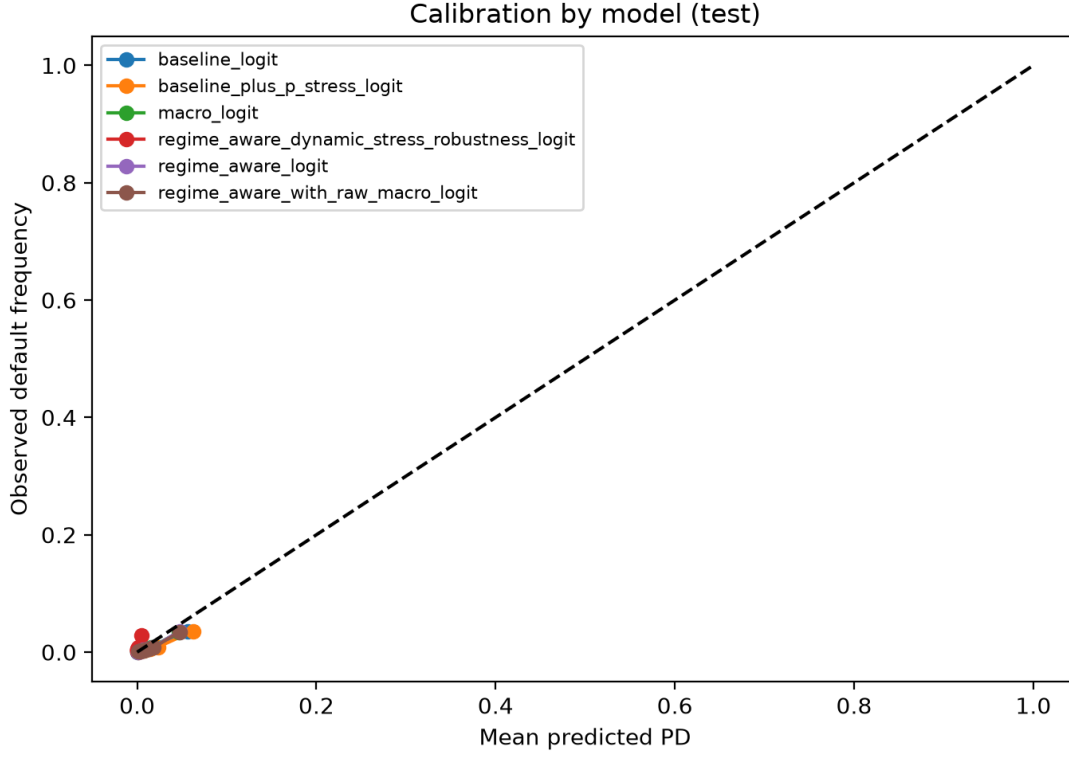


Figure 4.2: Calibration curves for the held-out test period. Points compare mean predicted PD with observed event frequency in equal-count probability bins.

Table 4.4: Calibration sensitivity for the principal test models

Model	Calibration	Mean PD	Brier	Log loss
Borrower-only	Raw	0.035379	0.008941	0.052552
Borrower-only	Intercept-only	0.011309	0.006120	0.033404
Borrower-only	Isotonic	0.011024	0.006125	0.033322
Borrower-only	Platt	0.010747	0.006055	0.033106
Regime-aware	Raw	0.063093	0.015897	0.084550
Regime-aware	Intercept-only	0.013791	0.006673	0.035555
Regime-aware	Isotonic	0.011020	0.006007	0.032811
Regime-aware	Platt	0.010802	0.006015	0.032945

4.6 Temporal, Portfolio, and Discrimination Diagnostics

The regime-aware model had a lower Brier score than borrower-only in both test-year groups. In 2023, the scores were 0.005432 and 0.005445. In 2024, they were 0.006546 and 0.006609. Both groups belonged to the same hard-stress classification and therefore did not constitute separate regime tests.

Monthly count calibration favoured borrower-only. Across the 18 test months, its mean absolute predicted-default-count error was 952.5, compared with 965.3 for regime-aware and 1,212.3 for macro-augmented. Mean absolute relative error was 0.6097 for borrower-only, 0.6367 for regime-aware, and 0.9016 for macro-augmented. The pooled Brier improvement did not translate into a lower average monthly count error.

Table 4.5: Monthly default-count error over the 18-month test period

Model	Mean absolute count error	Mean absolute relative error
Borrower-only	952.5	0.6097
Regime-aware	965.3	0.6367
Macro-augmented	1,212.3	0.9016
Dynamic stress	1,342.3	0.8751
Regime-aware plus macro	1,347.5	0.9957
Stress level only	1,546.0	1.0075

Mean predicted PD exceeded the event rate for the three core models, so the positive monthly count errors were consistent with the pooled mean-calibration result. Borrower-only produced the smallest average error even though regime-aware allocated risk more effectively across individual loan-months according to the pooled Brier and ranking measures. This difference separates cross-sectional allocation from portfolio-level calibration.

Figure 4.3 shows test ROC curves. The regime-aware ROC-AUC of 0.827256 exceeded the borrower-only value of 0.821979 and the macro-augmented value of 0.805984. PR-AUC showed the same ordering. This indicates better held-out ranking for the regime-aware model

in the terminal period. It does not resolve the mean overprediction or the weaker validation results.

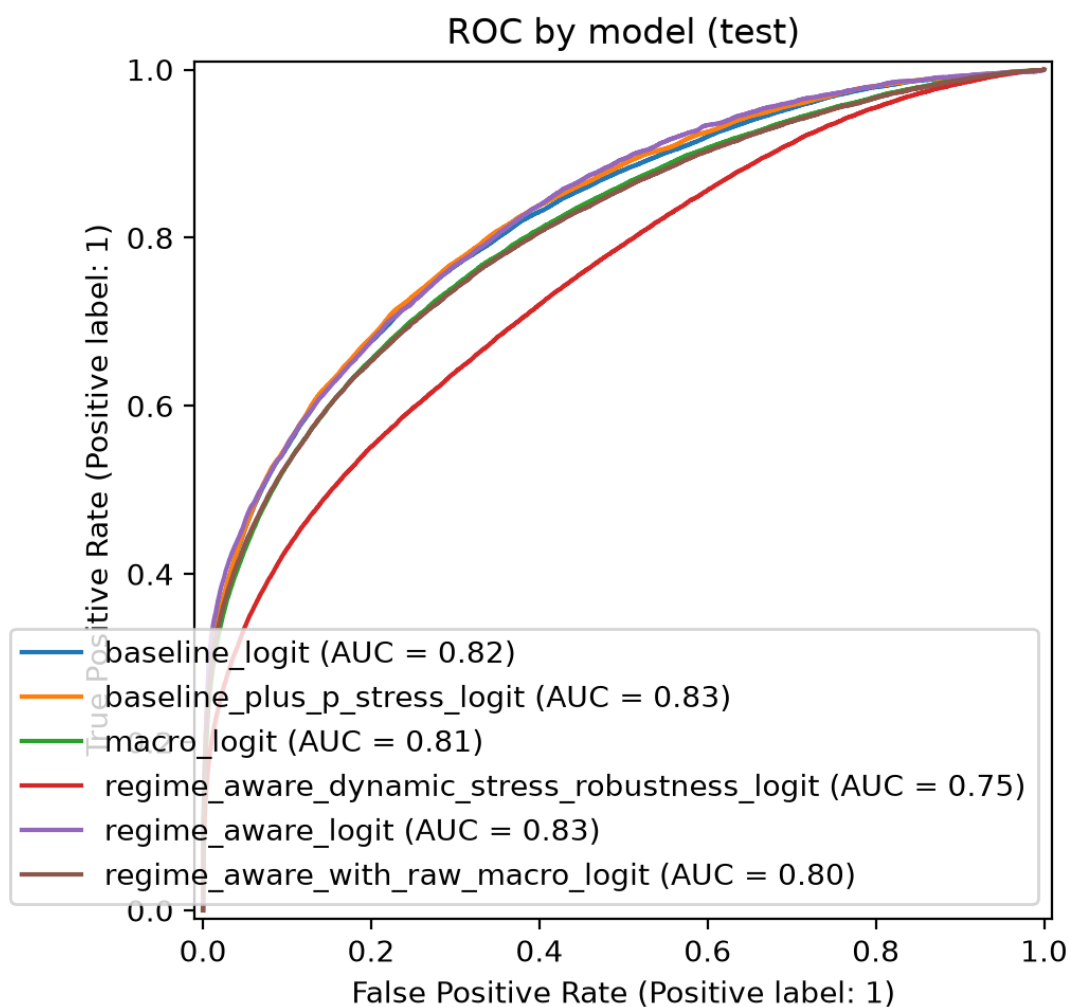


Figure 4.3: Receiver operating characteristic curves for the held-out test period.

Figure 4.4 reports the precision–recall curves. The test event rate of 0.006571 is the no-skill prevalence reference for average precision. The regime-aware PR-AUC was 0.160097, compared with 0.148361 for borrower-only and 0.133367 for macro-augmented. The ordering matched ROC-AUC, but the PR scale made the rare-event context explicit.

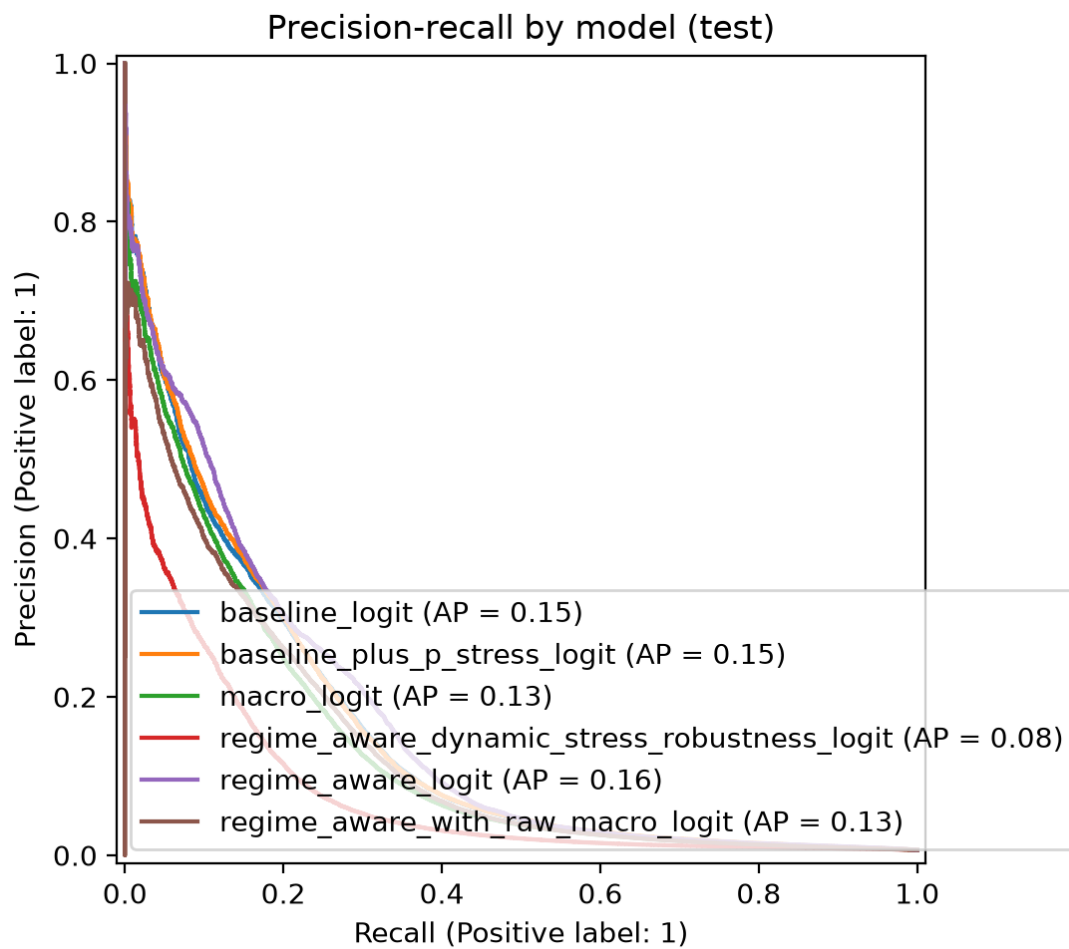


Figure 4.4: Precision-recall curves for the held-out test period.

4.7 Overall Interpretation

The regime-aware model answered the second research question through an explicit q_t level effect and eight borrower–regime interactions. The retained specification remained a regularised logistic model and did not include the raw macroeconomic vector. Its form allowed selected borrower sensitivities to change with the filtered stress probability.

For the third research question, the regime-aware model performed best on the pooled test Brier score, log loss, ROC-AUC, and PR-AUC. Borrower-only performed best on the corresponding validation measures and on monthly test default-count error. The macro-augmented baseline did not improve on borrower-only in the held-out test. The evidence is therefore mixed. It supports a modest test-period advantage for the regime-aware interaction model, subject to temporal instability and incomplete mean calibration.

The empirical design did not include rolling-origin HMM re-estimation or uncertainty intervals for differences between models. The test period also contained no hard-normal months. These limitations prevent a claim that the observed differences generalise across macroeconomic regimes or estimation windows.

The full-window HMM limitation and the test-period saturation are related but distinct. Full-window estimation affects the information available when the factor transformations and HMM parameters were fitted. Saturation affects the amount of variation in the resulting q_t during evaluation. Forward filtering addressed future factor observations in the state recursion. It did not address either parameter re-estimation at each forecast origin or the lack of a hard-normal test period.

The absence of resampled uncertainty is also material. Loan-month observations from the same loan are dependent, and all loans in a calendar month share the same macroeconomic input. A conventional interval that treated 4,106,275 test rows as independent would therefore be misleading. The retained artefacts support exact point comparisons only. The results distinguish their numerical evidence from broader claims about repeated samples or other economic periods.

Bibliography

- Bangia, A., Diebold, F. X., Kronimus, A., Schagen, C., and Schuermann, T. (2002). Ratings migration and the business cycle, with application to credit portfolio stress testing. *Journal of Banking & Finance*, 26(2–3):445–474.
- Basel Committee on Banking Supervision (2004). QIS 3: Definition of default. Basel II Quantitative Impact Study.
- Baum, L. E., Petrie, T., Soules, G., and Weiss, N. (1970). A maximization technique occurring in the statistical analysis of probabilistic functions of markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171.
- Bellotti, T. and Crook, J. (2009). Credit scoring with macroeconomic variables using survival analysis. *Journal of the Operational Research Society*, 60(12):1699–1707.
- Bellotti, T. and Crook, J. (2013). Retail credit stress testing using a discrete hazard model with macroeconomic factors. *Journal of the Operational Research Society*, 65(3):340–350.
- Bilmes, J. A. (1998). A gentle tutorial of the EM algorithm and its application to parameter estimation for Gaussian mixture and hidden markov models. Technical Report TR-97-021, International Computer Science Institute, Berkeley, CA.
- Board of Governors of the Federal Reserve System (2006). Basel ii: Definition of default.
- Brier, G. W. (1950). Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3.

- Costa e Silva, E., Lopes, I. C., Correia, A., and Faria, S. (2020). A logistic regression model for consumer default risk. *Journal of Applied Statistics*, 47(13–15):2879–2894.
- Davis, J. and Goadrich, M. (2006). The relationship between precision-recall and ROC curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240.
- Dimitriadis, T., Gneiting, T., and Jordan, A. I. (2020). Evaluating probabilistic classifiers: Reliability diagrams and score decompositions revisited.
- Federal Reserve Bank of St. Louis (2026). FRED economic data. Monthly macroeconomic data series.
- Freddie Mac (2026). Single-family loan-level dataset. Public loan-level origination and monthly performance data.
- Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.
- Hand, D. J. and Henley, W. E. (1997). Statistical classification methods in consumer credit scoring: A review. *Journal of the Royal Statistical Society: Series A*, 160(3):523–541.
- Murphy, A. H. (1973). A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600.
- Rabiner, L. R. (1989). A tutorial on hidden markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286.
- Saito, T. and Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, 10(3):e0118432.

- Thomas, L. C. (2000). A survey of credit and behavioural scoring: Forecasting financial risk of lending to consumers. *International Journal of Forecasting*, 16(2):149–172.
- Van Calster, B., McLernon, D. J., van Smeden, M., Wynants, L., and Steyerberg, E. W. (2019). Calibration: The achilles heel of predictive analytics. *BMC Medicine*, 17(1):230.
- Yu, F.-H., Lu, J., Gu, J.-W., and Ching, W.-K. (2018). Modeling credit risk with hidden markov default intensity. *Computational Economics*, 54(3):1213–1229.